# THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

## CLASS VII — SCIENCE · ANNUAL / FULL-YEAR PRACTICE PAPER

Session 2026-27 · Set 02 · Emphasis: Physical Science – Electricity, Heat, Motion, Light, Earth-Moon-Sun

| Class: 7 | Subject: Science | Set 02 | Time: 2 Hours 30 Minutes | Max. Marks: 60 |
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**General Instructions:**

1. All questions are compulsory.
2. This question paper has five sections A, B, C, D and E. Marks are indicated against each question.
3. Section A (Q. 1, parts I–XII) consists of multiple choice questions of 1 mark each.
4. Section B (Q. 2–5) consists of very short answer questions of 2 marks each.
5. Section C (Q. 6–8) consists of short answer questions of 3 marks each.
6. Section D (Q. 9–11) consists of long answer questions of 5 marks each.
7. Section E (Q. 12–15) consists of four case-based / source-based questions of 4 marks each (1 + 1 + 2).
8. There is no overall choice. However, internal choice is given in some questions. Only one of the choices in such questions is to be attempted.
9. Draw neat and labelled diagrams wherever required.

### SECTION A — Multiple Choice Questions — 1 questions · 12 marks

*Question 1 has twelve parts (I–XII). Each carries 1 mark.*

**1. From Question I–XII, choose the correct option.**

I. Which of the following substances turns blue litmus paper red? 1

(a) Vinegar (b) Lime water
(c) Soap solution (d) Baking soda solution

II. In an LED, the longer leg is connected to the: 1

(a) either terminal; it does not matter (b) switch only
(c) positive terminal of the cell (d) negative terminal of the cell

III. A metal that gives a ringing sound when struck is called: 1

(a) brittle (b) malleable (c) ductile (d) sonorous

IV. Which of the following changes can be reversed easily? 1

(a) Melting of ice (b) Burning of paper
(c) Rusting of iron (d) Ripening of a mango

V. Deficiency of iodine in the diet may cause the enlargement of which gland, called goitre? 1

(a) Pituitary (b) Adrenal (c) Pancreas (d) Thyroid

VI. The range of a clinical thermometer is: 1

(a) −10 °C to 110 °C (b) 0 °C to 100 °C
(c) 35 °C to 42 °C (d) 20 °C to 60 °C

VII. A bike moves at 72 km/h. Its speed in m/s is: 1

(a) 72 m/s (b) 15 m/s (c) 25 m/s (d) 20 m/s

VIII. Which organs of the human body remove nitrogenous wastes from the blood? 1

(a) Lungs (b) Liver only (c) Kidneys (d) Stomach

IX. Water and minerals taken up by roots are carried to the leaves through the: 1

(a) stomata (b) phloem (c) cuticle (d) xylem

X. A ray of light strikes a plane mirror at an angle of incidence of 35°. The angle of reflection is: 1

(a) 35° (b) 70° (c) 145° (d) 55°

XI. The Earth takes about 365¼ days to complete one revolution around the Sun. To adjust for the ¼ day, a leap year of ___ days is added once in four years. 1

(a) 364 (b) 366 (c) 367 (d) 365

XII A simple pendulum completes 20 oscillations in 40 s. Its time period is: 1

(a) 20 s (b) 60 s (c) 2 s (d) 0.5 s

## SECTION B — Very Short Answer Questions — 4 questions · 8 marks

*2 marks each (about 30–40 words).*

**2.** Ritu connected a cell, a bulb and a switch in a circuit but the bulb did not glow even after she closed the switch. Give any two possible reasons. **2**

**3.** Why does an antacid tablet give relief from acidity (indigestion)? Name the type of substance it contains. **2**

**4.** Convert the speed 54 km/h to m/s. How long will a bus moving at this speed take to travel 300 m? **2**

**OR**

Define the time period of a simple pendulum. Mention two ways in which the time period can be found more accurately.

**5.** Explain with an activity or a daily-life example how a shadow is formed. Why is a shadow darker than its surroundings? **2**

## SECTION C — Short Answer Questions — 3 questions · 9 marks

*3 marks each (about 60 words).*

**6.** Compare a plane mirror, a concave mirror and a convex mirror by giving one use of each. **3**

**7.** Explain why we have different seasons on the Earth. Why does the length of day vary? **3**

**OR**

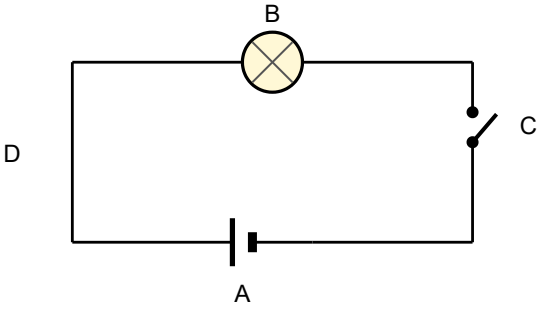
Distinguish between a solar eclipse and a lunar eclipse (any three points).

**8.** Explain the formation of a sea breeze. Why does it blow during the day? **3**

## SECTION D — Long Answer Questions — 3 questions · 15 marks

*5 marks each.*

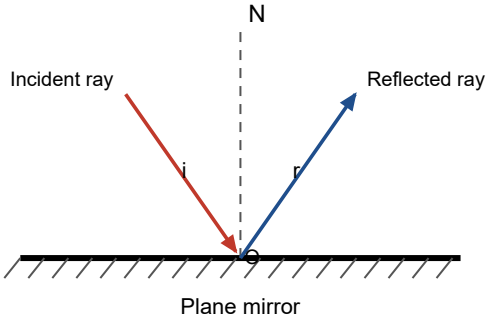
**9.** Study the circuit shown. (i) Name the parts A, B, C and D. (ii) Will the bulb glow? Give a reason. (iii) What should be done to make it glow? **5**




**OR**

(i) How does the heating effect of current help in an electric geyser? (ii) Why is the wire in a geyser made of nichrome? (iii) Why should we not touch bare electric wires? (iv) Differentiate between a conductor and an insulator with two examples each.

**10.** The figure shows a ray striking a plane mirror. (i) Name the normal and mark the angles i and r. (ii) Find the angle of reflection if the angle of incidence is 35°. (iii) What is the angle between the incident ray and the reflected ray? (iv) What is the angle between the incident ray and the mirror surface? (v) State the second law of reflection. **5**




**11.** (a) Explain the phases of the Moon. (3) **5**
(b) How many days does the Moon take to complete one cycle of phases (new moon to new moon)? (1)
(c) Name the phase when the whole of the Moon's lit side faces the Earth. (1)

**OR**

A train 200 m long moving at 90 km/h crosses a pole. (i) Express its speed in m/s. (ii) Find the time taken to cross the pole. (iii) A motorist travels 200 km at 80 km/h; find the time taken. (iv) What is the formula for speed? (v) Give two units of speed.

## SECTION E — Case-based / Source-based Questions — 4 questions · 16 marks

*Read the cases carefully and answer. Each question carries 4 marks.*

**12.** Aman's mother notices that the lights flicker whenever the geyser, iron and air-conditioner are switched on together in their house. The electrician says that the circuit is overloaded and advises installing an MCB.

(a) Which effect of electric current is used in an electric iron? 1

(b) Why does an overloaded circuit become dangerous? 1

(c) How does an MCB protect the house wiring? 2

**OR**

(d) Mention two safety rules while using electrical appliances.

**13.** In winter Nisha wears a woollen sweater and feels warm. At home she notes that the metal spoon in a hot cup of tea quickly becomes too hot to hold, while a wooden spoon does not.

(a) Which mode of heat transfer heats the metal spoon? 1

(b) Name one good and one poor conductor of heat from the passage. 1

(c) Why does a woollen sweater keep us warm? 2

**OR**

(d) Why do we wear white or light-coloured clothes in summer?

**14.** A rusted iron gate was painted by a painter. Another gate, made by a fabricator, was coated with zinc (galvanised). After two rainy seasons, the painted gate has again shown rust at some scratched places, but the galvanised gate remained shiny.

(a) Name the two substances necessary for rusting. 1

(b) Is rusting a physical or a chemical change? 1

(c) Why did the painted gate rust at scratched places? 2

**OR**

(d) Name another method of preventing rusting and one metal that does not rust easily.

**15.** After a meal, a cow's food travels through its digestive tract while the farmer's blood carries oxygen to every cell. Anita learns in class that animals take food, digest it, take in oxygen and send out wastes.

(a) Name the organ in humans where most of the water from undigested food is absorbed. 1

(b) Which blood vessels carry blood away from the heart? 1

(c) Explain the role of the kidneys in the human body. 2

**OR**

(d) Name the part through which the earthworm and frog partly breathe.